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(19) **United States**(12) **Patent Application Publication**
KALLENBACH et al.(10) **Pub. No.: US 2025/0283260 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SEWING UNIT****D05B 65/02** (2006.01)**D05B 75/00** (2006.01)(71) Applicant: **PFAFF Industriesysteme und Maschinen GmbH**, Kaiserslautern (DE)(52) **U.S. CL.**CPC **D05B 19/12** (2013.01); **D05B 57/26** (2013.01); **D05B 65/02** (2013.01); **D05B 75/00** (2013.01)(72) Inventors: **Harald KALLENBACH**, Heltersberg (DE); **Stefan SAMSTAG**, Lorsch (DE)(73) Assignee: **PFAFF Industriesysteme und Maschinen GmbH**, Kaiserslautern (DE)

(57)

ABSTRACT(21) Appl. No.: **18/861,429**(22) PCT Filed: **Jun. 13, 2023**(86) PCT No.: **PCT/EP2023/065759**

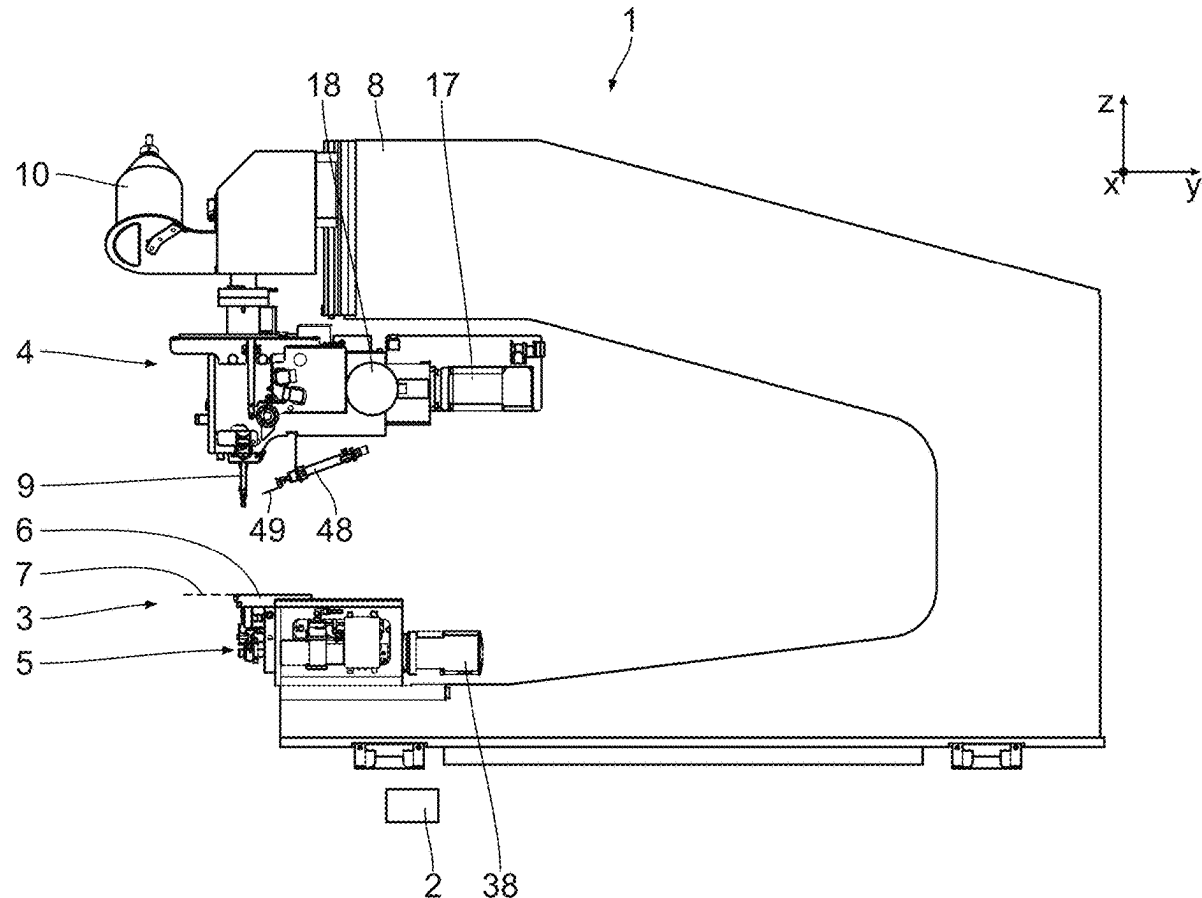
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A sewing unit has a CNC controller and a sewing machine. The sewing machine has a workpiece bed plate, an upper sewing part and a lower sewing part. The lower sewing part has a looper with a lower thread supply in a looper bobbin case. The looper can rotate about an axis which extends parallel to the workpiece bed. A bobbin case holding component disposed in the region of the workpiece bed is in engagement with the looper bobbin case such that it does not co-rotate the looper bobbin case during a rotation of the looper. The lower sewing part has a bobbin case lifter to displace the looper bobbin case between a neutral position and an opened position in which a free space for a thread between the looper bobbin case and the bobbin case holding component is enlarged in comparison with the neutral position.



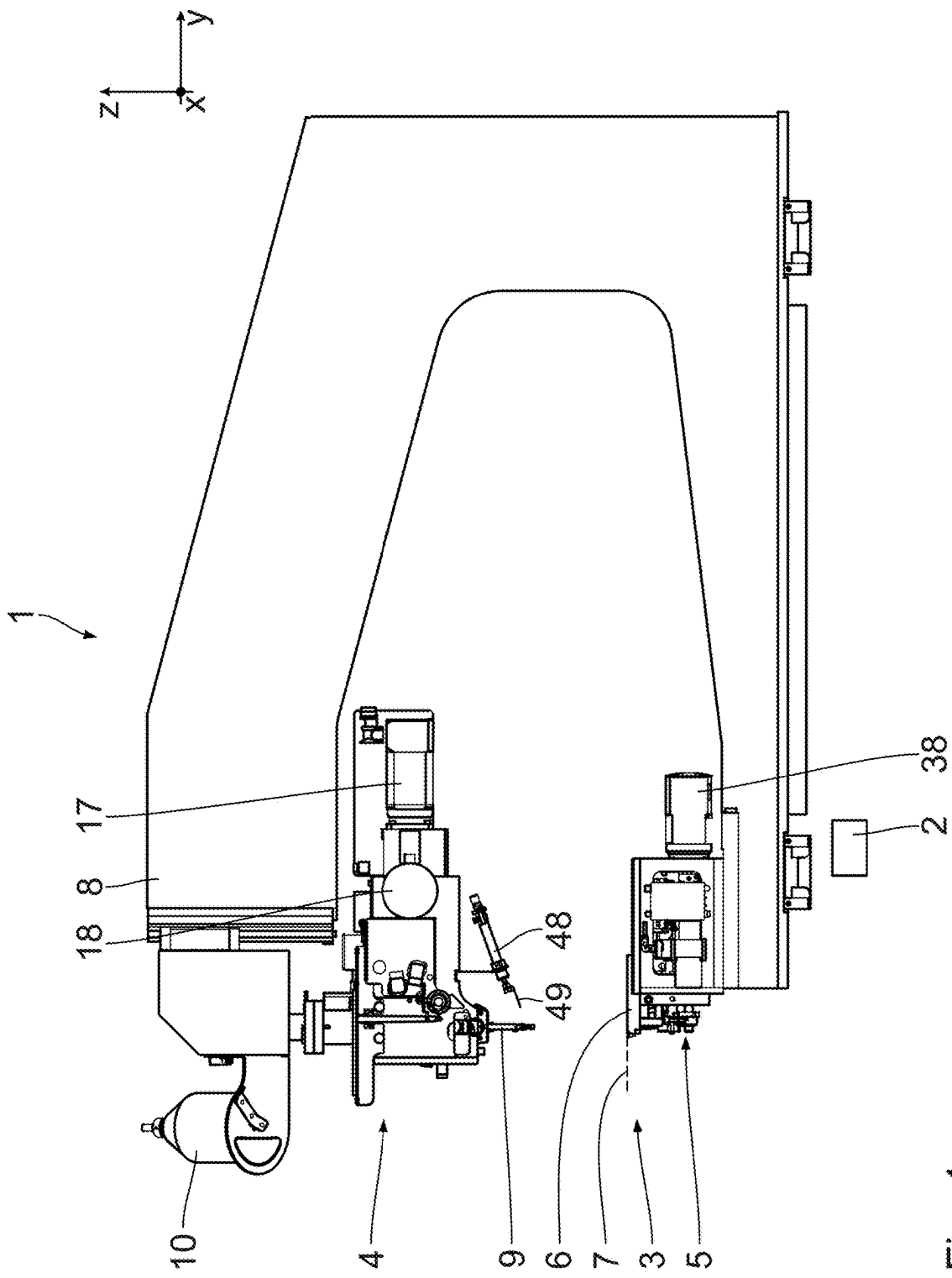


Fig. 1

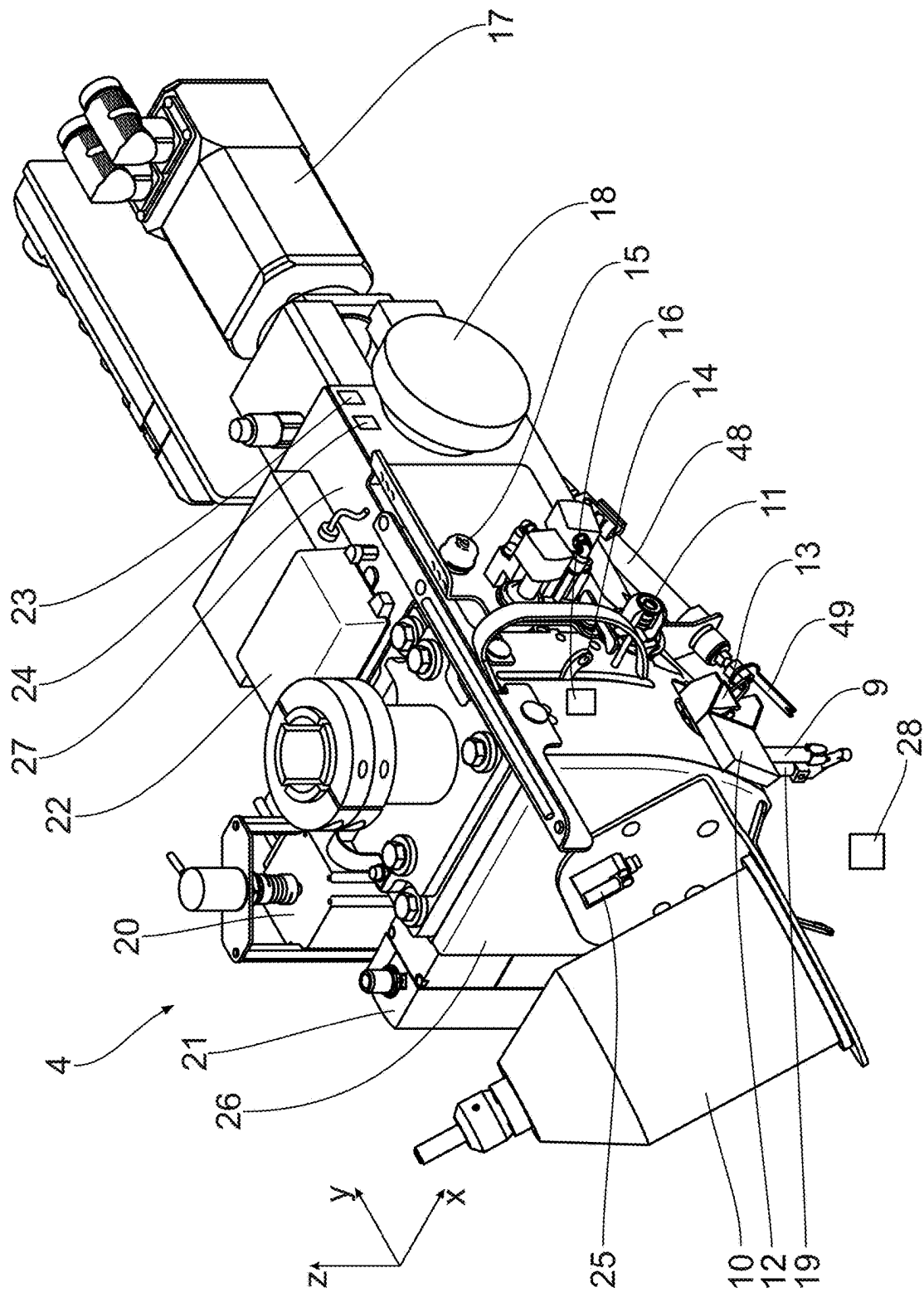


Fig. 2

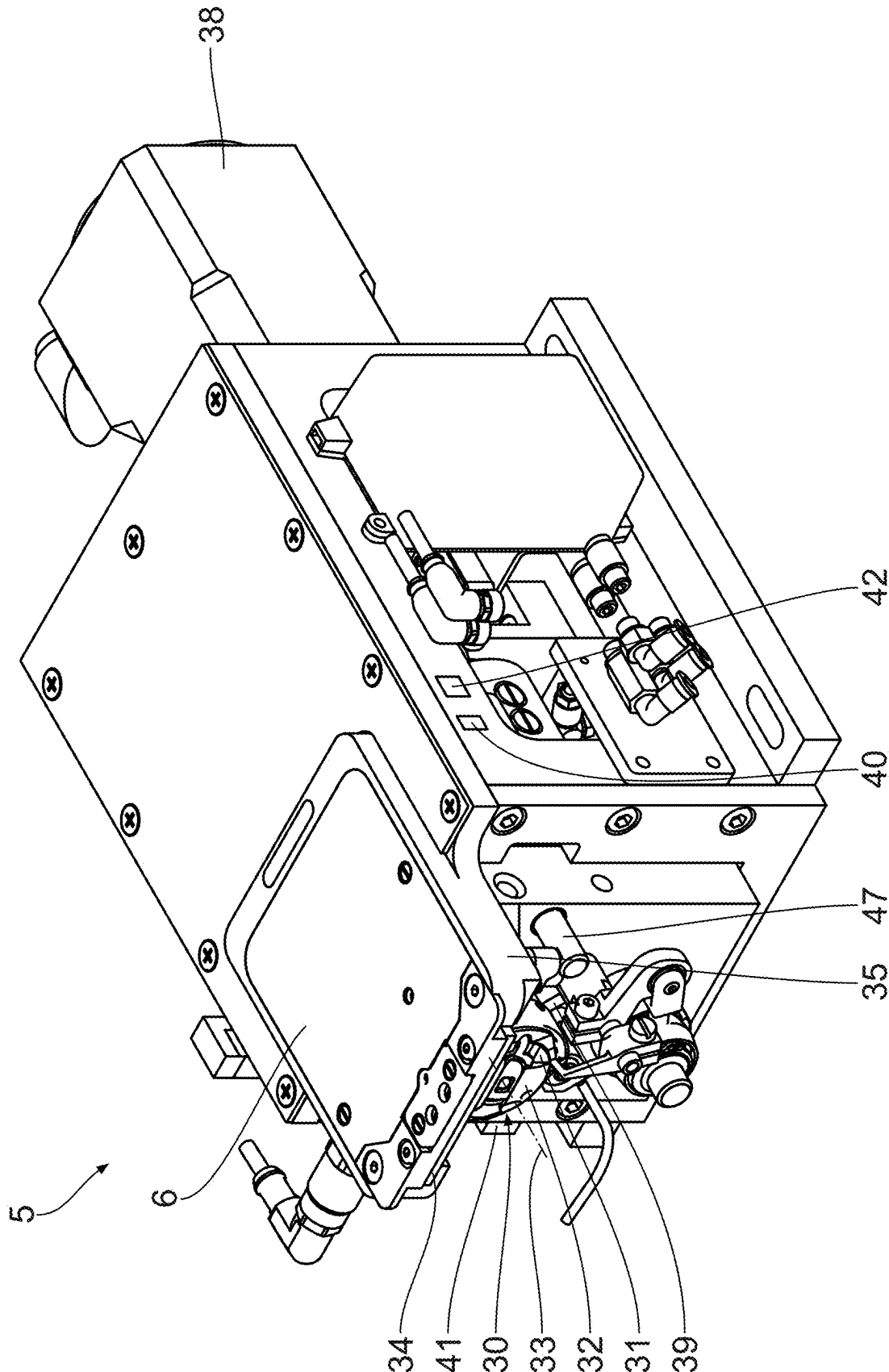


Fig. 3

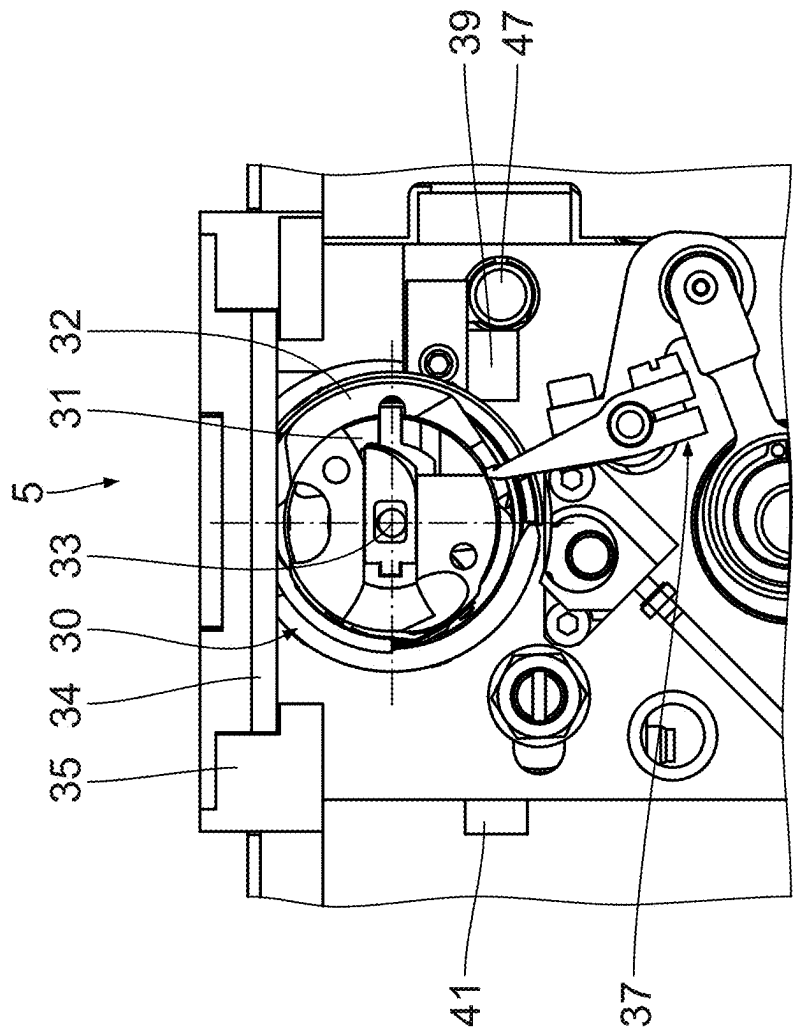


Fig. 4

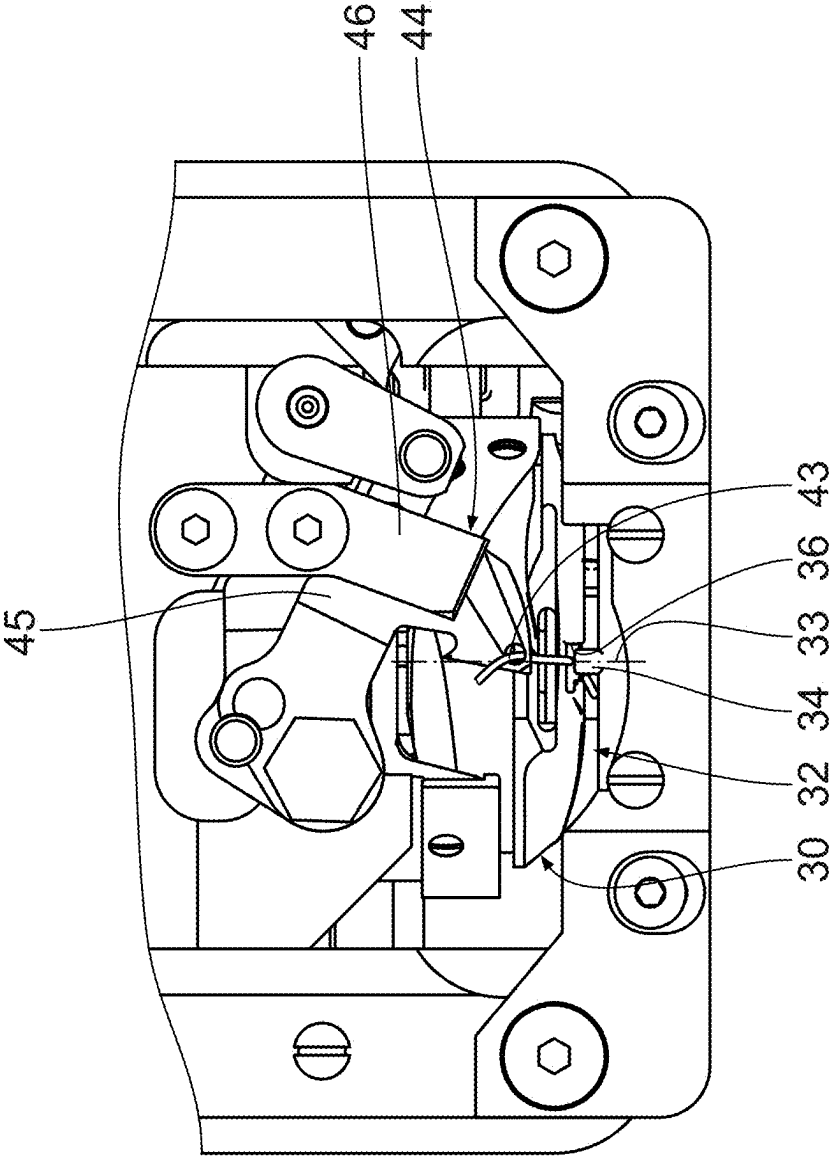


Fig.5

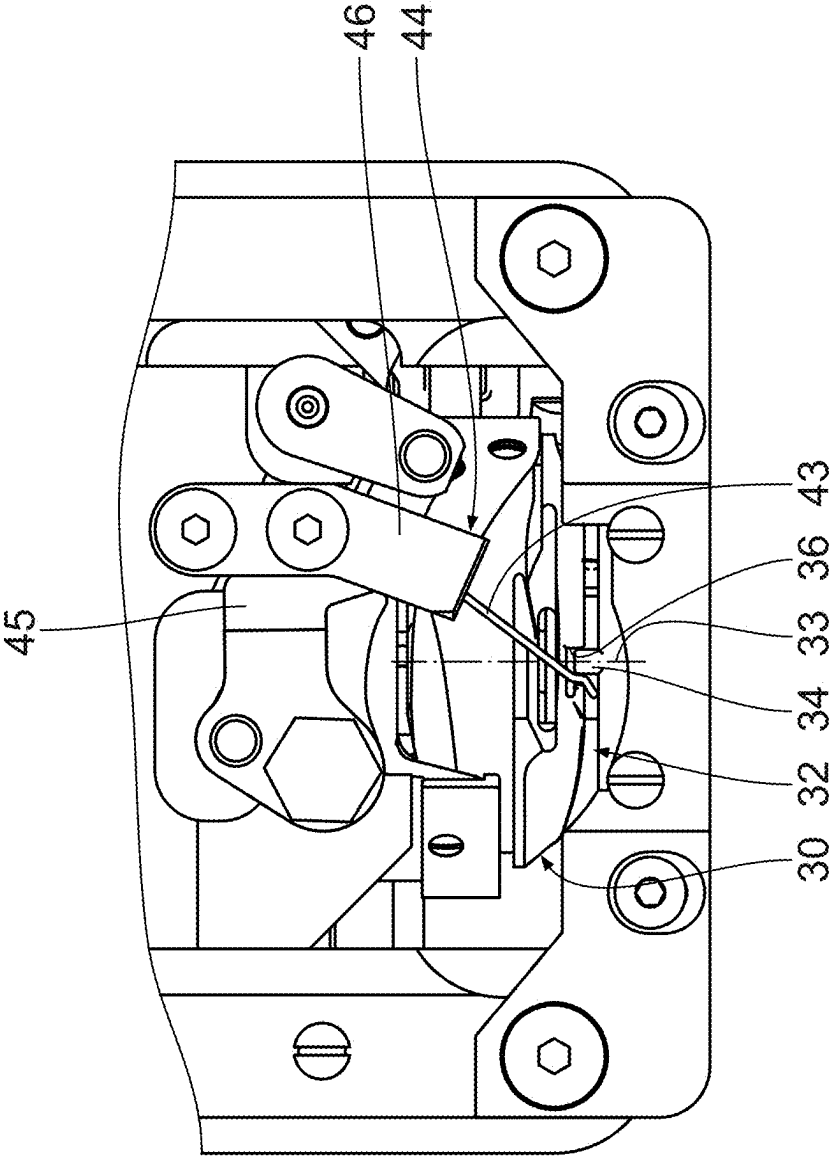


Fig.6

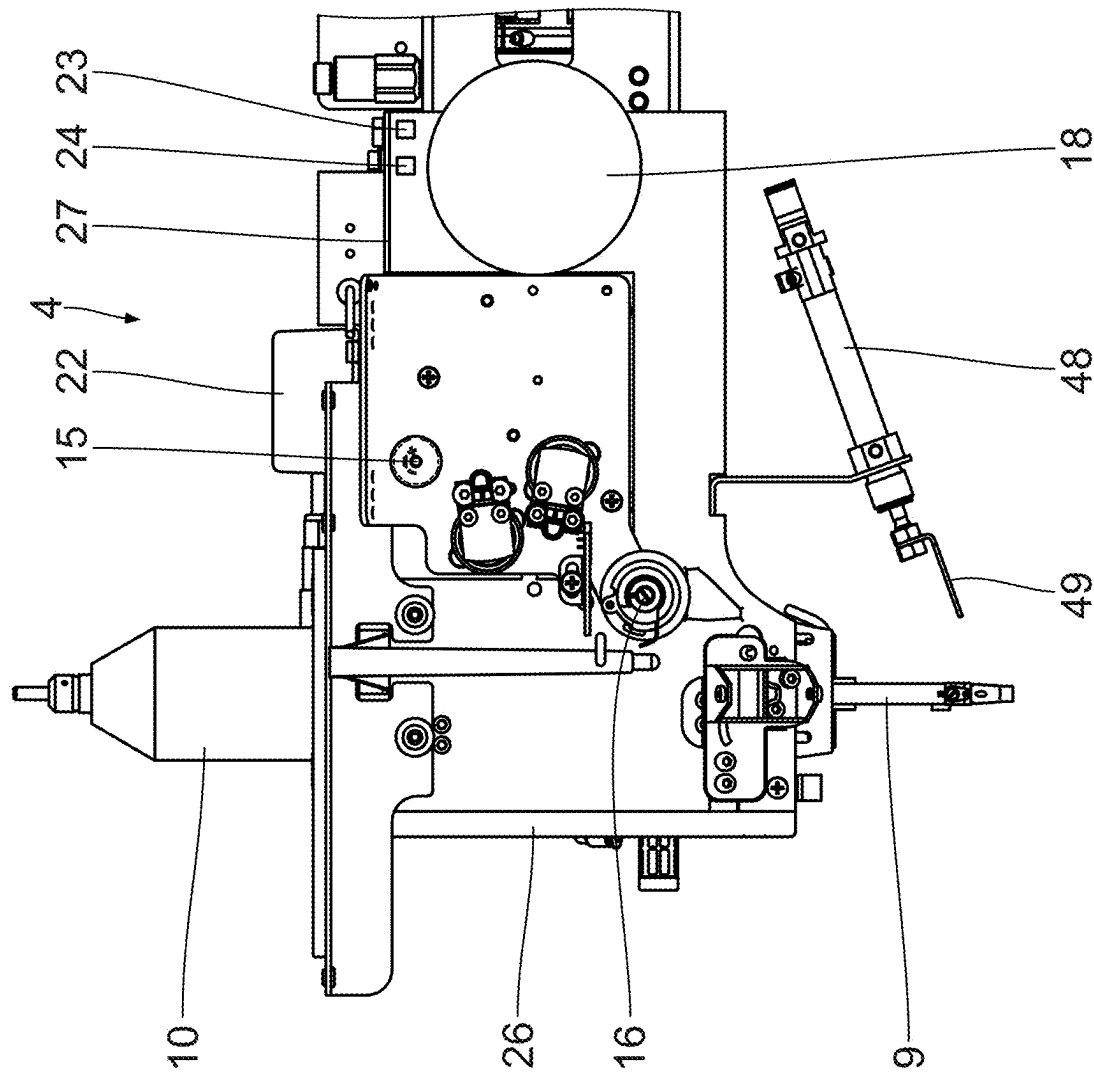


Fig. 7

SEWING UNIT

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a national stage application, filed under 35 U.S.C. § 371, of International Patent Application PCT/EP2023/065759, filed on Jun. 13, 2023, which claims the benefit of German Patent Application DE 10 2022 206 102.0, filed on Jun. 17, 2022, the content of which is herein incorporated by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a sewing unit.

BACKGROUND

[0003] Sewing units of this type are known commercially for the production of seams, e.g. for the automobile industry and in the field of technical textiles. EP 2 050 853 A1 discloses a sewing machine and a looper system for a sewing machine. DE 10 2018 205 835 A1 discloses a sewing machine. DE 41 28 336 C1 discloses a sewing machine for a stationary workpiece with a large surface area. DE 2 247 050 A discloses a thread take-up lever for sewing machines, preferably for multi-needle double lockstitch machines.

SUMMARY

[0004] An object of the present disclosure is to develop a sewing unit in such a way that mass production is thereby ensured with a high level of overall unit efficiency even when thick threads are being used.

[0005] This object is achieved by a sewing unit as described and claimed.

[0006] The inventors recognized that it is possible to modify a bobbin case lifter (bobbin case opener) known e.g. from DE 100 25 851 C1 and DE 32 44 946 C1 in such a way that, even with a very large looper, it can cooperate with correspondingly large lower thread supply. An example of a large looper is known from DE 20 2013 005 112 U1. The free space for the thread—which is enlarged by the bobbin case lifter—between the looper bobbin case and the bobbin case holding component reduces thread friction during sewing operation in the looper-side workpiece bed area. This reduced thread friction makes it possible to use even thick threads of NM 20 or even less. Such thick upper and/or lower threads can then be sewn at a sewing speed of more than 2000 stitches per minute, e.g. at 2000 stitches per minute, at 2200 stitches per minute, at 2500 stitches per minute or even at 2800 stitches per minute.

[0007] The lower sewing part can be designed in such a way that a semi-automatic or fully automatic bobbin exchanging system can be used for the correspondingly automated exchange of the lower thread supply. This additionally increases the efficiency of the sewing unit as a whole.

[0008] The sewing unit can be used to produce seams for the automobile industry, e.g. dashboard, vehicle seat, belt or air bag seams.

[0009] A C-frame or even another frame geometry can be used to hold the upper sewing part and the lower sewing part of the sewing machine of the sewing unit.

[0010] An up and down movement of the needle bar can take place along a z-coordinate of a sewing unit coordinate system. In this case, the upper sewing part jointly with the

lower sewing part and/or the workpiece bed plate can be designed to be displaceable along the x-coordinate and/or along the y-coordinate by means of correspondingly controlled drives. The sewing unit can thus be designed with a movable sewing machine and/or with a movable workpiece bed plate.

[0011] The sewing machine can be designed with monitoring of a residual thread quantity for the upper thread and/or the lower thread. The sewing machine can be designed with at least one vibration sensor for monitoring a maintenance requirement, in particular of bearings of the sewing unit, in particular of the upper sewing part and/or of the lower sewing part. The upper sewing part can be designed with needle cooling.

[0012] A thread cutting device for cutting off thread ends of the upper thread and/or of a lower thread which protrude above a formed seam and have a maximum excess length of 15 mm leads to neater seam formation. A thread cutting device like the one known from EP 2 300 654 B1 can be used. An excess length achieved with the thread cutting device can amount to 15 mm but can also be less, e.g. 12 mm or 10 mm. A starting thread and/or an end thread of the formed seam can be cut off. A thread finger can cooperate with the thread cutting device when the thread is being cut. A corresponding process can include, after a stop in sewing, thread capture of the thread to be cut with the thread finger for pulling the thread, and subsequently cutting of the thread captured after the thread has been pulled by the thread cutting device.

[0013] An embodiment of the thread cutting device which comprises a cutting blade which is driven in a controlled manner by a dual-acting pneumatic cylinder has proved its worth in the generation of the shortest possible excess length within a reproducible process. The cutting blade can be guided by means of a cam or track.

[0014] A narrow housing head design for the upper sewing part wherein the upper sewing part comprises a housing arm portion, wherein the housing arm portion terminates in a housing head, wherein the needle bar is guided in the housing head, and wherein the housing head in one operational embodiment including the upper thread supply has a width in a dimension parallel to the workpiece bed and perpendicular to the housing arm portion which amounts to a maximum of 300 mm makes it possible for a plurality of sewing machines within the sewing unit to cooperate next to each other in parallel in order to work across a large sewing area. The width of the housing head of the upper sewing part can amount to 280 mm, 250 mm and can even be less. This width is generally at least 200 mm.

[0015] Mounting of the upper thread supply on the housing head as an extension of the housing arm portion makes possible a particularly narrow width for the upper sewing part in the operational embodiment. The upper thread supply can then be designed to be aligned with the housing arm portion.

[0016] A thread tensioner device for setting a thread tension of the upper thread makes possible good seam formation control. The thread tension can be adjustable in different ways for individual seam sections of a whole seam to be produced.

[0017] A presser foot allows an adaptation of the sewing process to different workpiece conditions. The presser foot adjustability can be possible individually for seam sections

of a whole seam to be produced, and so corresponding section-dependent requirements can be taken into account.

[0018] Exemplified embodiments of the invention will be explained in greater detail hereinunder with reference to the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 schematically illustrates a side view of main components of a CNC-controlled sewing unit with a sewing machine, comprising an upper sewing part and a lower sewing part cooperating therewith;

[0020] FIG. 2 is a perspective view of an upper sewing part for use in the sewing unit of FIG. 1;

[0021] FIG. 3 is a perspective view of a lower sewing part for use in the sewing unit of FIG. 1;

[0022] FIG. 4 is an end view of details of the lower sewing part including a bobbin case lifter which can be driven in a controlled manner;

[0023] FIG. 5 shows, in a view of the lower sewing part obliquely from above, a looper bobbin case of a looper relative to a bobbin case holding component in a neutral position;

[0024] FIG. 6 shows, in an illustration similar to FIG. 5, the looper bobbin case in a position which is opened in comparison to the neutral position by reason of an effect of a bobbin case lifter, which position is also referred to as the open position; and

[0025] FIG. 7 shows, in a side view, an embodiment of the upper sewing part with a thread finger for pulling back an upper thread after sewing a seam and before thread cutting.

DETAILED DESCRIPTION

[0026] A sewing unit 1 has a CNC controller 2, illustrated schematically in FIG. 1, and a sewing machine 3. The sewing machine 3 has an upper sewing part 4 and a lower sewing part 5.

[0027] The sewing unit 1 further has a sewing bed plate 6 which provides a workpiece bed in a workpiece bed plane 7.

[0028] The upper sewing part 4 and the lower sewing part 5 of the sewing machine 3 are mounted on a frame 8 of the sewing machine 3 of the sewing unit 1.

[0029] A Cartesian xyz-coordinate system is used hereinunder to make it easier to describe positional relationships. In FIG. 1, the x-axis extends perpendicularly to the plane of the drawing and out of this plane. The y-axis extends horizontally to the right in FIG. 1. The z-axis extends vertically upwards in FIG. 1.

[0030] The frame 8 is roughly C-shaped in such a way that a usable sewing surface in the xy-plane and in particular also along the y-direction can extend over a large area which can be greater than 1000 mm×1000 mm.

[0031] During sewing, the workpiece bed plate 6 is displaced relative to the upper sewing part 4 on the one hand and to the lower sewing part 5 on the other hand along the x-direction and/or along the y-direction. Depending on the design of the sewing unit 1, this relative displacement can be such that the frame 8 of the sewing machine 3 moves relative to the workpiece bed plate 6, the workpiece bed plate 6 is thus fixed, or such that the workpiece bed plate 6 moves relative to the then stationary frame 8. In a further variant, both the frame 8 and also the workpiece bed plate 6 can move relative to each other along the x-direction and/or along the y-direction.

[0032] FIG. 2 shows details of the upper sewing part 4. The upper sewing part 4 has a needle bar 9 which can move up and down in a controlled manner along the z-direction. Depending on the design of the upper sewing part 4, the needle bar 9 for guiding a sewing needle held thereon, not shown in FIG. 2, can be cooled.

[0033] The upper sewing part 4 has an upper thread guide for guiding an upper thread from an upper thread supply 10 in the form of a thread reel to the needle bar 9. The upper thread is not shown along this upper thread guide in FIG. 2. A thread gripper 11 which is disposed close to the needle bar 9 and which is activated by means of a gripper drive 12, which can be of pneumatic design, appertains to this upper thread guide.

[0034] In the region of the thread gripper 11, a thread break sensor 13, which can be designed as an optical sensor, can be disposed. By means of such a thread break sensor 13, it is possible to monitor whether upper thread is still being supplied to the needle bar 9.

[0035] A thread take-up lever 14 and a thread tension adjusting device 15, with which thread tension of the upper thread can be set, also appertain to the upper thread guide. Depending on the design of the upper sewing part 4, in the region of the thread take-up lever 14, a thread tension sensor 16 can be disposed, by means of which thread tension of the upper thread can be monitored.

[0036] The movement of the needle bar 9 and of the thread take-up lever 14 is driven in a controlled manner by means of a sewing drive motor 17. Sewing tools of the sewing machine 3 can be made to approach a position manually by means of a hand wheel 18 of the upper sewing part 4. An axis of rotation of the hand wheel 18 extends parallel to the x-direction, i.e. perpendicularly to a C-plane (yz-plane) of the frame 8 and also perpendicularly to the extension of an arm shaft (extension along the y-direction) of the upper sewing part 4, from the rotational movement of which the driving of the needle bar 9 and of the thread take-up lever 14 is derived.

[0037] The upper sewing part 4 can additionally be fitted with a presser foot, not shown in FIG. 2. Such a presser foot is held on a presser foot bar 19. An adjusting drive 20, which is designed as a servomotor or pneumatic drive, is used for height adjustment of the presser foot bar 19 along the z-direction.

[0038] A pneumatic fan drive 21 can be provided to give resilience to the presser foot

[0039] bar 19.

[0040] A further monitoring sensor 22 of the upper sewing part 4 monitors an "upper dead centre" position of the needle bar 9. The monitoring sensor 22 can thus be designed as an optical sensor.

[0041] In order to monitor a bearing condition of the upper sewing part 4, this part can comprise a vibration sensor 23. As soon as vibrations of the upper sewing part 4 sensed by the vibration sensor 23 exceed a tolerance value, a signal can be output to prepare for maintenance of the sewing machine 3.

[0042] The upper sewing part 4 has a fill level sensor 24 in order to monitor the lubricant level, in particular an oil level, for lubrication of moving parts of the upper sewing part 4.

[0043] An upper thread fill level sensor 25, which can be an optical sensor, is used to monitor a fill level of the upper thread supply 10.

[0044] The upper thread supply 10 is mounted on a housing head 26 of the upper sewing part 4. This mounting position of the upper thread supply 10 is in an extension of a housing arm portion 27 of the upper sewing part 4 which extends along the y-direction and terminates in the housing head 26.

[0045] The needle bar 9 and the presser foot bar 19 are guided in the housing head 26

[0046] Depending on the design of the upper sewing part 4, the housing head 26 in one operational embodiment including the upper thread supply 10 can have a width in the x-dimension, i.e. parallel to the workpiece bed plane 7 and perpendicular to the extension dimension y of the housing arm portion 27, which amounts to a maximum of 300 mm and which can amount to e.g. 280 mm or even 250 mm.

[0047] Alternatively, the upper thread supply 10 can also be mounted on the frame 8 or even disposed next to the housing head 26, as shown in FIGS. 1 and 7.

[0048] Furthermore, a reading device 28—illustrated schematically in FIG. 2—for an upper thread type, which is provided with the upper thread supply 10, appertains to the sewing unit 1. The reading device 28 can be e.g. a bar code reader.

[0049] FIG. 3 shows an embodiment of the lower sewing part 5 of the sewing machine 3 of the sewing unit 1. The lower sewing part 5 cooperates with the upper sewing part 4 in order to form a seam. For this purpose, the lower sewing part 5 has a looper 30 which can be rotationally driven in a controlled manner and has a lower thread supply in a looper bobbin 31 which is housed in a looper bobbin case 32. The looper 30 has a looper point which cooperates with the sewing needle fastened to the needle bar 9 of the upper sewing part 4 in order to form stitches or a seam. In this way, a lockstitch, in particular a double lockstitch can be produced.

[0050] Including the looper point, the looper 30 can be rotated about an axis of rotation 33 extending parallel to the y-axis. This axis of rotation 33 extends parallel to the workpiece bed 7, i.e. parallel to the xy-plane.

[0051] The receiving area in the looper bobbin case 32 for the looper bobbin 31 with the lower thread supply has an outer diameter, measured perpendicularly to the axis of rotation 33, which is greater than 30 mm.

[0052] In the region of the workpiece bed plate 6 there is disposed a holding web 34 as a bobbin case holding component (cf. FIGS. 5 and 6) which is fixedly connected to a frame 35 of the lower sewing part 5. The holding web 34 is in engagement with a holding aperture 36 of the looper bobbin case 32 in such a way that, during a rotation of the looper 30 about the axis of rotation 33, the looper bobbin case 32 does not rotate therewith. In the circumferential direction about the axis of rotation 33, the holding aperture 36 is more extensive than the holding web 34 and so there are gaps, i.e. a free space, in this circumferential direction, depending on the circumferential position of the holding aperture 36 relative to the holding web 34.

[0053] The lower sewing part 5 has a bobbin case lifter (bobbin case opener) 37 which can be driven in a controlled manner (cf. FIG. 4) for displacement of the looper bobbin case 32 in the circumferential direction about the axis of rotation 33 between a neutral position, which is illustrated in FIG. 5, and a lifted (opened) position illustrated in FIG. 6.

[0054] A lever point of the bobbin case lifter 37 cooperates with a counter surface of the looper bobbin case 32 in order to displace the looper bobbin case 32 in the circumferential direction.

[0055] The rotational driving of the looper 30 and also the driving of the bobbin case lifter 37 are derived from a sewing drive 38 of the lower sewing part 5.

[0056] A stitch length during seam formation can be in the range between 2 mm and 10 mm.

[0057] The lower sewing part 5 in turn has a lower thread fill level sensor 39, a monitoring sensor 20 for monitoring an initial position of a looper point of the looper 30, a vibration sensor 41 for monitoring a bearing condition of bearings of the lower sewing part 5 and a lubricant level sensor 42, which correspond in their structure and manner of operation to the sensors already explained above with reference to the upper sewing part 4.

[0058] In the opened position of the looper bobbin case 32 of FIG. 6, this position is shifted relative to the holding web 34 in an anti-clockwise direction in comparison with the position of FIG. 5 by a few angular degrees in the circumferential direction about the axis of rotation 33 in FIGS. 4 to 6. On the side of the holding web 34 which is on the left in FIG. 6, in the opened position a free space is then produced, which is enlarged in comparison with the neutral position, between the holding web 34 and the holding aperture 36 of the looper bobbin case 32, this free space being for passage of a lower thread 43 between the looper bobbin case 32 and the bed-side components, in particular the holding web 34.

[0059] In the opened position of FIG. 6, this enlarged free space reduces thread friction during sewing operation of the sewing machine 3. Even thick threads 43 with a weight of at least 50 g per 1000 m of thread (50 tex=20 NM) can be sewn at an operating speed of the sewing machine 3 of at least 2000 stitches per minute. The operating speed can be e.g. 2200, 2400 or even 2500 stitches per minute and can be even higher. The thread can also be thicker or heavier than 50 tex, e.g. 77 tex (=13 NM).

[0060] The upper thread and/or lower thread can be a polyamide thread and/or a nylon thread. The upper thread and the lower thread can be of different colours.

[0061] The looper 30 is attached to the frame 35 of the lower sewing part 5 in such a way that, in particular, an automatic exchange system for exchanging the looper bobbin 31, i.e. of the lower thread supply, is rendered possible.

[0062] With the sewing unit 1, seams can be produced for the automobile industry, e.g. dashboard, vehicle seat, belt or air bag seams.

[0063] In addition, the lower sewing part 5 has a thread cutting device 44, components of which are illustrated in FIGS. 5 and 6. The thread cutting device 44 is used to cut off thread ends of the upper thread or of the lower thread which protrude above a seam formed by the sewing machine 3. After the thread cutting device 44 has cut them off, cut-off thread ends remain which have an excess length above the formed seam of at most 15 mm. This excess length can also be shorter and e.g. amount to 12 mm or 10 mm. The thread ends can be a starting thread or an end thread of the seam.

[0064] The thread cutting device 44 has a cutting blade 45, the thread catching hook and the blade thereof are concealed in FIGS. 5 and 6 by a counter blade 46 of the thread cutting device 44. The cutting blade 45 is shown in a cutting position in FIG. 5 and in a neutral position in FIG. 6. In order to cut thread, the cutting blade 45 is pivoted about a pivot

axis—extending parallel to the z-axis—in the direction towards the thread run in the region of the holding aperture 36, wherein the thread catching hook captures the thread or threads to be cut. When returning to the neutral position, the captured thread or captured threads are cut off between the cutting edge of the cutting blade 45 and the counter blade 46.

[0065] A blade drive is used for controlled driving of the cutting blade 45 and acts upon a drive shaft 47, cf. FIG. 3, which is a dual-acting pneumatic cylinder. The movement of the cutting blade 45 is guided by means of a cam or track not shown in the drawing.

[0066] Depending on the design of the sewing machine 3, this machine additionally has a thread finger device 48 with a thread finger 49 (cf. FIG. 7) which cooperates with the thread cutting device 44 in order to cut thread.

[0067] FIG. 7 shows an embodiment of such a thread finger device 48. This is mounted on the upper sewing part 4. The thread finger 49 can be displaced using a pneumatic drive between the neutral position of FIG. 7 and a cutting preparation position. In the cutting preparation position, after a stop in sewing by the sewing machine 3, the thread finger 49 pulls the thread to be cut back into a position in which it can be captured by the cutting blade 45. After this pulling back of the thread to be cut by means of the thread finger 49, the thread is then cut by the thread cutting device 44.

[0068] Depending on the design, the sewing unit 1 can also comprise several pairs made up in each case of an upper sewing part 4 and a lower sewing part 5 cooperating therewith to form a seam, i.e. a plurality of sewing machines like the sewing machine 3 explained above. This plurality of sewing machines can each be mounted on an associated frame like the frame 8. The various frames 8 of this plurality of sewing machines 3 can be displaced relative to each other along the x-direction and/or along the y-direction by means of respective individually activatable displacement drives. In particular, in the case of very large sewing areas to be sewn, a plurality of sewing machines 3 can then work on the sewing area in parallel. This parallel working is controlled by means of the CNC controller 2 of the sewing unit 1.

[0069] Both thread tension adjustment by means of the thread tension adjuster 15 and also foot height adjustment of the presser foot by means of the adjusting drive 20 can be individually adjustable in a controlled manner for each seam section by means of the CNC controller 2. This can take into account sewing requirements which differ depending on the section, e.g. section-dependent sewing of different workpiece layer numbers and/or different workpiece thicknesses and/or different stitch geometries.

[0070] The CNC controller 2 is in signal communication with the various drives and the sensors of the sewing unit 1, as explained above. For this purpose, the sewing unit 1 has a corresponding data bus system. Signal transmission can be effected by means of a real time Ethernet. An example of this is the Ether CAT protocol.

[0071] By means of such an Ethernet system, signal data are relayed to a production management system of the sewing unit 1 for the purpose of production control, which is also known as a manufacturing execution system (MES).

1-8. (canceled)

9. A sewing unit (1), comprising:

a CNC controller (2); and

at least one sewing machine (3), having

a workpiece bed plate (6) for provision of a workpiece bed (7),

an upper sewing part (4) with

a needle bar (9) which can move up and down in a controlled manner and

an upper thread guide for guiding an upper thread from an upper thread supply (10) to the needle bar (9), and

at least one lower sewing part (5), cooperating with the upper sewing part (4) to

form a seam,

with a looper (30),

wherein the looper (30) can be rotationally driven in a controlled manner and has a lower thread supply (31) in a looper bobbin case (32),

wherein the looper (30) can rotate about an axis (33) which extends parallel to the workpiece bed (7),

wherein the looper (30) has a diameter which amounts to at least 30 mm,

with a bobbin case holding component (34) disposed in a region of the workpiece bed (7) being in engagement with the looper bobbin case (32) in such a way that, during a rotation of the looper (30), the looper bobbin case (32) does not rotate therewith, and

with a bobbin case lifter (37) which can be driven in a controlled manner for displacement of the looper bobbin case (32) between

a neutral position and

an opened position in which a free space for a thread (43) between the looper bobbin case (32) and the bobbin case holding component (34) is enlarged in comparison with the neutral position.

10. The sewing unit as claimed in claim 9, further comprising a thread cutting device (44) for cutting off thread ends of the upper thread and/or of a lower thread which protrude above a formed seam and have a maximum excess length of 15 mm.

11. The sewing unit as claimed in claim 10, wherein the thread cutting device (44) comprises a cutting blade (45) which is driven in a controlled manner by a dual-acting pneumatic cylinder (47).

12. The sewing unit as claimed in claim 9, wherein the upper sewing part (4) comprises a housing arm portion (27),

wherein the housing arm portion (27) terminates in a housing head (26),

wherein the needle bar (9) is guided in the housing head (26),

wherein the housing head (26) in one operational embodiment including the upper thread supply (10) has a width in a dimension (x) parallel to the workpiece bed (7) and perpendicular to the housing arm portion (27) which amounts to a maximum of 300 mm.

13. The sewing unit as claimed in claim 12, wherein the upper thread supply (10) is mounted on the housing head (26) as an extension of the housing arm portion (27).

14. The sewing unit as claimed in claim 9, wherein the at least one sewing machine (3) comprises an activatable thread tensioner device (15) for setting a thread tension of the upper thread.

15. The sewing unit as claimed in claim 9, wherein the at least one sewing machine (3) comprises a presser foot bar (19) with activatable foot height adjustment (20).

16. The sewing unit as claimed in claim **9**, wherein the at least one sewing machine (**3**) is one of a plurality of sewing machines (**3**), and wherein the plurality of sewing machines (**3**) are supported by a common support frame of the sewing unit (**1**) and can be displaced with respect to each other along at least one displacement dimension in a controlled drivable manner.

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